Engine Model(s): PT6A

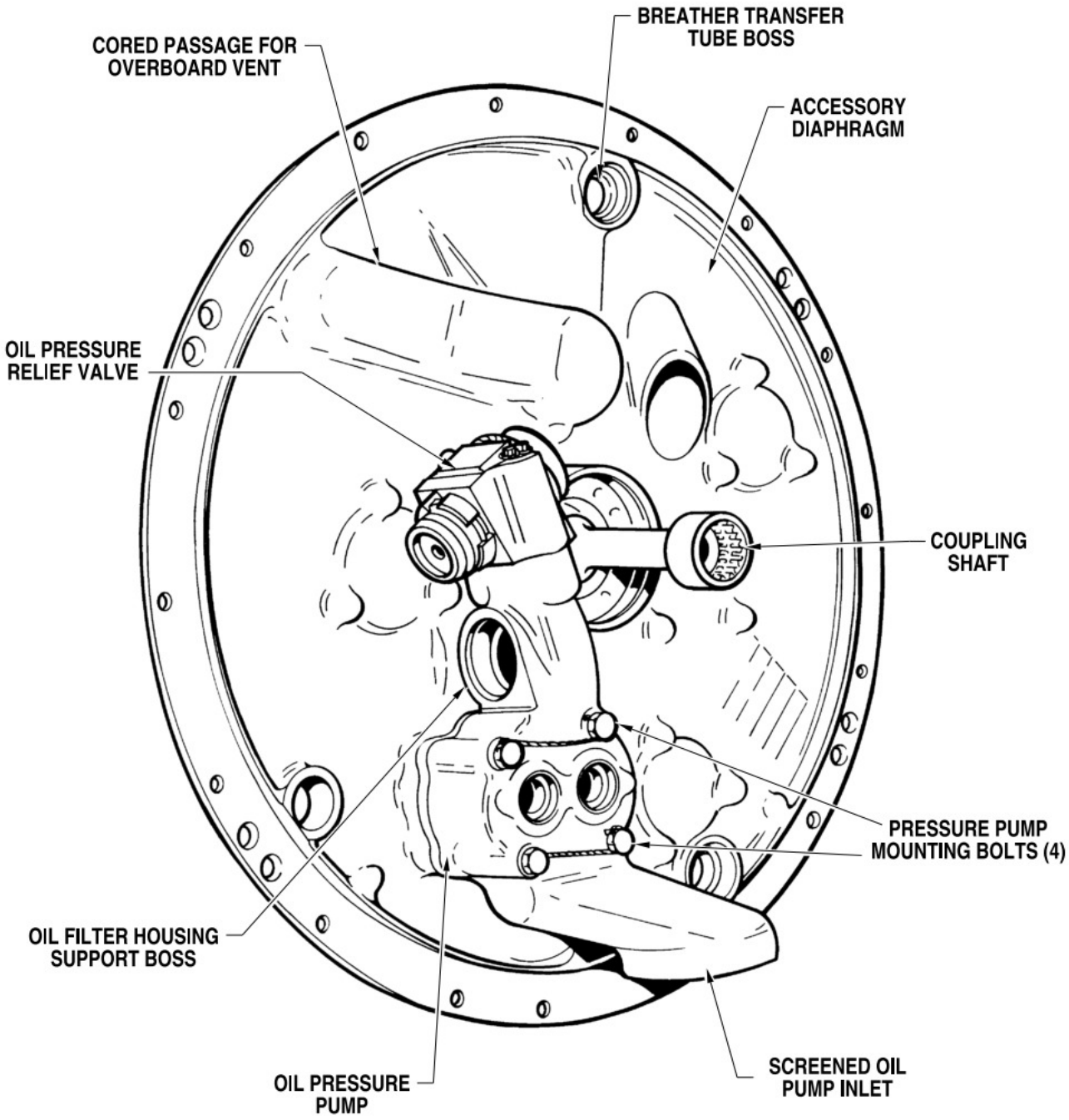
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OIL DISTRIBUTION - DESCRIPTION AND OPERATION

1. Description and Operation (Ref. Fig. 1)

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Figure 1 Accessory Gearbox Diaphragm and Oil Pump



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A. Oil Tank

The oil tank is an integral part of the compressor inlet case and is located in front of the accessory gearbox. The tank is provided with a filler neck and an integral quantity dipstick and cap located at the 11 o'clock position on the accessory gearbox housing. The cap and dipstick are fastened to the filler neck, which passes through the gearbox housing and accessory diaphragm and into the oil tank. The markings on the dipstick correspond to US quarts, and indicate the amount of oil required to fill the tank to the full mark. An anti-flooding device and breather arrangement, located at the 12 o'clock position on the compressor inlet case, prevents foaming when scavenged oil is returned to the tank during engine operation. The tank is vented to the top of the accessory gearbox, where the air/oil mist is passed through a centrifugal breather. The breather separates the oil from the air and returns the oil to the bottom of the accessory gearbox. The relatively oil free air is discharged overboard via a cored passage in the accessory diaphragm and transfer tube, to a breather boss located at the 2 o'clock position on the gearbox housing. Draining of the oil tank is done by removal of a plug located at the 6 o'clock position of the compressor inlet case (Ref. 72-00-00).

Refer to Chapter 72-00-00, DESCRIPTION AND OPERATION, Table 8 for oil tank data.

B. Oil Pump

Pressure oil is circulated from the integral oil tank and through the engine lubricating system by a self-contained gear-type pump mounted at the bottom of the oil tank. The oil pump consists of two gears contained in a cast housing bolted to the front of the AGB diaphragm. The pump gears are driven by an accessory gearshaft which also drives the internal scavenge pump. A removable inlet screen is fitted at the oil pump. The pump housing incorporates a support boss at the pump for the oil filter housing, and is counterbored to accommodate the check valve which is mounted at the inner end of the filter housing. The lubrication system pressure relief valve (Ref. 72-60-00) is installed on a pad located at the upper end of the housing. An internal cored passage connects the relief valve to pump outlet.

C. Oil Filter

The oil filter, installed in the pressure line downstream of the oil pump, consists of cartridge type filter element, a spring loaded bypass valve, and a check valve mounted in a removable alloy housing. Depending on installation requirements, the filter element may be of the cleanable (reusable) or non-cleanable (disposable) type.

D. Centrifugal Breather

The centrifugal breather consists of an aluminum alloy, shrouded impeller housing attached to the rear face of the starter generator gearshaft by a retaining ring. Rotational torque is transmitted from the gearshaft to the impeller by three equi-distant pins. Breather air flows radially inward through the rotating impeller housing where oil particles in the breather air are separated by centrifugal force, and drain freely into the gearbox sump. The relatively oil-free air passes inward and into the hollow rear section of the gearshaft. The air flows forward and into a cored passage in the accessory diaphragm. From the cored passage, the air passes through a transfer tube to a breather boss on the accessory gearbox housing at the 2 o'clock position.

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